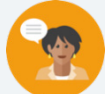


11.1 Adding Rational Numbers

Lesson Introduction

Lesson Introduction				
Benchmark	MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.		Learning Target	I can fluently add rational numbers.
Benchmark Coherence	Building On		Working Toward	
	MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.		MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.	
Essential Questions	- How is the process of adding integers apply to adding rational numbers? - How is the sign of the sum determined when adding rational numbers?			
Lesson Narrative	This lesson opens with a video about a landscape architect where collaboration is an apparent part her job. Next, the lesson presents adding decimals, fractions, and integers separately, as well as investigating the process of each. The lesson continues with adding mixed numbers and describing the process. Students practice adding rational numbers, and the lesson wraps up with students selecting their own fraction or mixed number and finding the sum.			
Component Summary	11.1.1 Warm-Up (~5 min.)	Career Profile Video (<i>SE p. 165</i>)	11.1.4 Refresher (15 min.)	Add and subtract rational numbers and complete statements (<i>SE p. 167</i>)
	11.1.2 Guided Instruction (10 min.)	Analysis of student work of adding decimals and fractions (<i>SE p. 166</i>)	11.1.5 Guided Practice (10 min.)	Complete addition expressions (<i>SE p. 169</i>)
	11.1.3 Try It (5 min.)	Pair equivalent rational expressions (<i>SE p. 167</i>)	11.1.6 Your Turn! (5-10 min.)	Find the sum of rational number expressions (<i>SE p. 170</i>)
	11.1.7 Wrap-Up (~5 min.)	Add rational numbers (<i>SE p. 170</i>)		
Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - N/A <u>Additional Terms:</u> - Absolute value		- Career Profile Video - Calculators (only for question 7 in 11.1.4)	11.1.1 Warm-Up contains a video to accompany the questions.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not convert the numerators after changing the denominators to the common denominator of fractions. - Students may not include the whole number when adding mixed numbers and fractions.	- Students may write the common denominator and the new numerators adjacent to the original fraction. Consider advising the students to rewrite the equivalent fractions underneath the original fractions to assist with organization and clarity. - Determine if your students need the support of all or part of the refresher. - If time is an issue, then consider assigning 11.1.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Compare and Connect 11.1.5 Guided Practice presents students with adding two fractions and then adding two mixed numbers. Consider using a compare and contrast routine where the students compare and contrast the differences and similarities between the processes for adding fractions and adding mixed numbers.	MA.K12.MTR.3.1: Mathematical Fluency Mathematical Fluency covers flexibility and precision while carrying out calculations. By providing students with multiple opportunities to execute procedures and perform calculations, this lesson allows students to maintain flexibility and accurately add a variety of rational numbers.
	11.1.3 Try It and 11.1.6 Your Turn! both provide practice opportunities for students to engage with a variety of tools and resources that provide differentiated support. Students can engage with On-Ramp videos on the following topics for additional support: - Adding Decimals - Add Fractions with Unlike Denominators	
Lesson Notes:		

11.1.1 Warm-Up: Landscape Architect

Component Overview: Play the Career Profile video on the Landscape Architect for the entire class. After students watch the video (3:18 minutes), prompt them to reflect on the profession by answering the two questions independently.



11.1.1 Warm-Up: Landscape Architect

Janice Nicol, a landscape architect, works to design natural systems and outdoor spaces, like trails and parks, for campuses and recreational facilities. Her job is both creative and technical. Creativity is expressed through drawing sketches both by hand and by using computer design programs and imaging. The technical side of her job involves knowing things such as land grading, water calculations, and natural systems.

Janice mentions that she regularly collaborates and works together with others while doing her job.

1. Why do you think it is important to be able to collaborate when doing a job?
Answers will vary. It is important to collaborate because different people have different ways of thinking that can help get the job done quicker.

2. How can you build your collaboration skills?
Answers will vary. You can build collaboration skills by asking others their opinions or asking what their thought process is on a specific thing.



In order to assist students with crafting complete responses, post a thinking map, such as a circle map, of key characteristics of a collaborative team member. Some of these characteristics can include being a good listener or a creative contributor toward team goals.

11.1.2 Guided Instruction: Adding Positive Rational Numbers

Component Overview: Students complete an error analysis of student work to determine the mistake in adding decimals and correct it. Students then add fractions with different denominators and complete statements to explain how to add fractions with unlike denominators.



11.1.2 Guided Instruction: Adding Positive Rational Numbers

1. Adrian added 5.25 and 6.4. His work is shown.

Adrian's Work	
5.25	
+ 6.4	
5.89	



Consider providing students with a place value chart to help align the decimals correctly to find the sum. This will provide a helpful visual for students to determine the error in Adrian's work.

- a. Write an explanation to Adrian detailing any mistakes he made and how he can correct his work.
Adrian, you did not line up your decimals to ensure the place values matched; this is necessary in order to add correctly.

- b. What is the sum of 5.25 and 6.4?
 11.65



When finding the sum of decimal numbers, add values with the same place value.

11.1.2 Guided Instruction: Adding Positive Rational Numbers (continued)

2. Johannes added $\frac{1}{8}$ and $\frac{3}{4}$. Part of her work is shown.

a. Complete Johannes' work by filling in the boxes.

Johannes' Answer
$\frac{1}{8} + \frac{3}{4} =$
$\frac{1}{8} + \frac{3}{4} \left(\frac{2}{2} \right) =$
$\frac{1}{8} + \frac{6}{8} = \frac{7}{8}$

b. Complete the statements by choosing a word or number from the bank. Not all words and numbers may be used, and some may be used more than once.

Word Bank		
2	4	8
numerators	common denominator	
one half	fourths	seven eighths

To find the sum of $\frac{1}{8}$ and $\frac{3}{4}$, Johannes first rewrote each fraction using a common denominator.

The common denominator of 8 and 4 that

Johannes used was 8. Then, she added the two

numerators together to find the total

number of seven eighths.



Prior to completing question 2b, encourage students to discuss the vocabulary in the Word Bank. Students may recall that the numerator is the number above the fraction bar in the fraction. Students might also point out that common denominators are the numbers below the fraction bar that are the same when adding two or more fractions.

11.1.3 Try It: Adding Positive Rational Numbers

Component Overview: Students will solve fraction expressions and will pair the expressions that result in the same sum.



11.1.3 Try It: Adding Positive Rational Numbers

1. Match pairs of expressions that have the same sum.
Write the equivalent expressions on the lines provided.

$\frac{3}{4} + \frac{4}{8}$	$\frac{2}{3} + \frac{3}{5}$	$\frac{2}{3} + \frac{7}{18}$
$\frac{5}{9} + \frac{1}{2}$	$\frac{15}{15} + \frac{8}{30}$	$\frac{12}{16} + \frac{1}{2}$

$$\underline{\frac{3}{4} + \frac{4}{8}} = \underline{\frac{12}{16} + \frac{1}{2}}$$

$$\underline{\frac{2}{3} + \frac{3}{5}} = \underline{\frac{15}{15} + \frac{8}{30}}$$

$$\underline{\frac{2}{3} + \frac{7}{18}} = \underline{\frac{5}{9} + \frac{1}{2}}$$



As you circulate listen for the strategies students use to determine their answers. Are they finding common denominators for each sum or are they simplifying fractions? Are they matching based on LCM's of the denominator? Invite students with different strategies to share them with their peers.



To provide students with a challenge, consider inquiring about one common denominator for all twelve fractions. This will give students the opportunity to find the least common denominator for a wide range of numbers.

11.1.4 Refresher: Operations with Integers

Component Overview: Students will work with a partner to find the sum of addition expressions and will answer questions regarding the process. Students will also verify solutions with a calculator and will complete statements about the process.



11.1.4 Refresher: Operations with Integers

1. Work with your partner to complete the table. The first addition expression has been done for you.

Expression	Absolute Value of First Addend	Absolute Value of Second Addend	Addend With Greater Absolute Value	Sign of Sum	Sum
$(-10) + 6$	10	6	-10	-	-4
$10 + (-6)$	10	6	10	+	4
$(-10) + (-6)$	10	6	-10	-	-16
$10 + 6$	10	6	10	+	16

2. How did the size of the absolute value for each addend determine the sign of the sum?
The sign of the addend with the greater absolute value is the sign that the sum will have.
3. Explain how finding the sum for the first two expressions was different from finding the sum for the last two expressions.
The first two expressions have addends with different signs; the answer is the difference between the absolute values of the addends and takes the sign of the addend with the larger absolute value. The last two expressions have addends with the same sign, and the answer is the sum of the absolute values of the addends and has the same sign as the addends.
4. How were the sums for $10 + (-6)$ and $(-10) + 6$ similar?
The absolute values of the sums are the same.



If students still struggle with operations with integers, then consider providing students with integer chips to assist them with completing the table for question 1. This manipulative will assist students with observing the sign of each integer.



Consider making connections to student responses to question 3 when completing statements in question 6. Students will likely provide statements similar to what is structured in question 6.

11.1.4 Refresher: Operations with Integers (continued)

5. How were the sums for $(-10) + (-6)$ and $10 + 6$ different?

The signs of the sums are different.

6. Complete the statements using the words *same* or *different* to describe the rules for finding the sums of signed numbers.

When two addends have the same sign, the sum will have the sign of the addends.

When two addends have different signs, the sum will have the sign of the addend with the greater absolute value.

7. Apply the rules for integers to these expressions with rational numbers. Find the sum of each expression. Then, verify your answer using a calculator.

Addition Expression	Sum When Using the Rule	Sum With a Calculator	Does the Rule Work?
$(-0.92) + 50.6$	49.68	49.68	Yes
$4.95 + 3.05$	8	8	Yes
$\left(-\frac{4}{5}\right) + \left(\frac{22}{25}\right)$	$\frac{2}{25}$	$\frac{2}{25}$	Yes
$1.25 + (-6.45)$	-5.20	-5.20	Yes
$\left(-\frac{2}{3}\right) + \left(-\frac{9}{20}\right)$	$-\frac{67}{60}$	$-\frac{67}{60}$	Yes
$(-17.01) + (-2.99)$	-20	-20	Yes

8. Complete the sentence.

The sign rules for adding integers are

- ☒ the same
☐ not the same

as the rules for adding rational numbers.



Prompt students to revisit their responses to question 3 when answering question 6.



Students are extending the rules they learned for operations with integers to the operations with rational numbers. Help students connect that the rules they already know are applied to rational numbers. Consider assigning each addition expression in the table to different sets of students.

Consider whether to verify the sums using a calculator as a whole group or whether to provide a calculator to certain students to verify. Calculators should not be used on any other part of the lesson.

Encourage the students to add on an additional column to the table to show their work. The space to provide work on the table will emphasize solving the questions using the rule first, instead of just using the calculator to find the sum.

11.1.5 Guided Practice: Applying Signed Number Rules to Rational Numbers

Component Overview: Students fill in boxes to make incomplete equations true. Students then collaborate with a partner to describe how to determine the missing value in equations.



11.1.5 Guided Practice: Applying Signed Number Rules to Rational Numbers

- The blanks in the addition equation represent missing values. Fill in the blanks for the addition equation below to make a true statement.

Answers will vary.

$$-12\frac{3}{4} + (-5\frac{\boxed{3}}{6}) = -18\frac{\boxed{3}}{12}$$

- Find the sum for each addition expression.

Addition Expression	Sum
$\frac{4}{5} + (-\frac{3}{7})$	$\frac{13}{35}$
$-9\frac{1}{2} + (-14\frac{1}{8})$	$-23\frac{5}{8}$



Prior to completing question 2, encourage students to compare and contrast the processes for adding and subtracting fractions versus the processes for adding and subtracting mixed numbers.

- The blanks in the addition equation represent digits in the values.

- Fill in the blanks for the addition equation below to create a true equation.

$$-16.537 + 6.1\boxed{7}8 = -10.\boxed{3}59$$

- Describe the process you used to determine the values for each of the blanks above.

I lined up the decimals to subtract. I figured out what the missing numbers were by finding numbers that would make the subtraction statement true.

- Discuss your process with your partner. Write a summary of their process. Have your partner initial to show that your summary accurately expressed their process.

Your Partner's Process

Answers will vary. I added 16.537 and $-10.\underline{\quad}59$. This gave me the missing 7 for 6.178. Then 3 was the only option for the blank in $-10.\underline{\quad}59$.

Partner's Initials SS

11.1.6 Your Turn!

Component Overview: Students will find the sum for each rational number expression.



11.1.6 Your Turn!

1. Find the sum for each expression.

a. $-1\frac{1}{2} + (-\frac{1}{8})$

$-1\frac{5}{8}$

b. $-12.37 + 18.161$

5.791

c. $-\frac{2}{5} + (-7\frac{1}{4})$

$-7\frac{13}{20}$

d. $-4.48 + 9.366$

4.886

e. $-233.24 + 25.8$

-207.44

f. $-4\frac{1}{7} + 17\frac{2}{9}$

$13\frac{5}{63}$



As you circulate or review student work look for common errors students are making. Consider whether that error is based on their understanding of adding signed numbers, fractions, mixed numbers, or place value. Then determine any necessary small group intervention. Fluency with rational numbers is an important focus in Grade 6 Accelerated. Students will continue to build on their current understanding throughout the year.



Provide students with blank place value charts. Allow students to fill in the charts with the given decimal numbers in order to assist them with lining up decimals and the place values.



For students who need additional support with either fractions or decimals, provide On-Ramp video and practice tools to supplement their learning.

11.1.7 Wrap-Up: Ticket Out the Door

Component Overview: Fractions and mixed numbers are given in two rows. Students will choose one number from each row with the intent to find a negative answer and will find the sum. Students should not use a calculator to complete the Wrap-Up.



11.1.7 Wrap-Up: Ticket Out the Door

- Choose a value from Row A and a value from Row B that, when added, will result in a negative answer. Then, calculate the sum.

Answers will vary.

Row A

$-1\frac{3}{5}$	$-\frac{1}{8}$	$-2\frac{1}{4}$
-----------------	----------------	-----------------

Row B

$1\frac{1}{2}$	$2\frac{6}{25}$	$\frac{19}{40}$
----------------	-----------------	-----------------

Row A

$-2\frac{1}{4}$

+

Row B

$1\frac{1}{2}$

=

Sum

$-\frac{3}{4}$



Monitor for the approach that each student takes, whether they find common denominators first or round and estimate based on approximations.



If time permits, challenge students to determine how many different possible sums they could find and order their approximations from least to greatest.